**Faculty of Applied Sciences and Technology**
**Semester II Session 2022/23**

# Assignment

*Submit on/ before 28th June 2023*

Create an animated 2D projectile simulator in MATLAB. The main purpose of this simulator is to find out whether a projectile can hit an obstacle or not. Assume that the projectile is a particle launched at height $y_0$, angle $\theta$ at velocity $v_0$. Then the displacement coordinates of the projectile at time $t$ is

$$x = v_0 t \cos\theta$$
$$y = y_0 + v_0 t \sin\theta - \frac{1}{2}gt^2$$

where $g$ is the gravity acceleration. Let the obstacle be a rectangular object with height $h$, width $w$ and placed at $l$ units away from the origin. Include the following features:

1. The user can change the values of $y_0$, $\theta$, $v_0$, $g$, $h$, $w$ and $l$. The simulator calculates and animates the trajectory of the projectile.
2. Stop the projectile once it hits the obstacle.
3. Display the current position of the projectile and the current time in a text box. The position and the time changes in real time while the projectile is moving.

In addition, let the user compare the trajectory between 2 projectiles. For example, the user can see how different the trajectories can be by modifying the launching angle of the projectiles while setting the other parameters to be the same.

Your application should answer these questions

1. when the user changes the value of the velocity, display what is the minimum and the maximum velocity needed to clear the obstacle.
2. when the user changes the value of the angle, display what is the minimum and the maximum angle needed to clear the obstacle.

The animation should be precise, smooth and runs at reasonable speed. Use your creativity to include more useful features in your application.